SAR Image Reconstruction Method for Target Detection Using Self-Attention CNN-Based Deep Prior Learning

Min Li, *Student Member, IEEE*, Weibo Huo, *Member, IEEE*, Junjie Wu, *Senior Member, IEEE*, and Jianyu Yang, *Member, IEEE*

***Abstract*— Due to its all-day and all-weather capability, synthetic aperture radar (SAR) plays an important role in many remote sensing and monitoring applications. However, conventional SAR image reconstruction methods generally perform undifferentiated imaging, complicating target detection. To address this challenge, we propose a SAR image reconstruction method based on self-attention deep prior learning for differentiated image reconstruction and target detection. The proposed method can separate the target from the clutter using their feature priors during image reconstruction, thus helping improve target detection performance. Specifically, a deep prior learning operation based on a self-attention convolutional neural network (SACNN) is proposed. SACNN can help enhance target and suppress clutter by learning both the local and global features. Finally, the proposed method is implemented through an unrolled deep network with a loss function designed to make the reconstructed image beneficial for target detection. Simulation experiments have been conducted to verify the efficacy of the proposed method.**

***Index Terms*— Deep prior, self-attention convolutional neural network (SACNN), synthetic aperture radar (SAR) image reconstruction, target enhancement, unrolling algorithm.**

I. Introduction

SYNTHETIC aperture radar (SAR) has the ability to obtain high-resolution electromagnetic scattering images of the observed scene in all-day and all-weather conditions. Target detection is an important application of SAR and is widely adopted in maritime search and rescue, target surveillance, etc. Therefore, the studies of SAR image reconstruction and target detection have attracted much attention. In conventional SAR processing, image reconstruction and target detection are usually considered as two independent processes. Typically, the SAR image is obtained from the sampled echo data with an image reconstruction method. Then, the SAR image target detection method locates the region of interest by exploiting specific target features, such as the intensity feature, texture feature, and structure feature [1]. The SAR image reconstruction method will determine the available target features, which in turn affects the subsequent target detection performance. Hence, the target detection performance can be improved by enhancing the extraction of the target features during the image reconstruction process.

In general, the conventional SAR image reconstruction methods, e.g., backward projection (BP), range-Doppler (RD), and chirp scaling algorithm (CSA), were developed to achieve the accurate estimation of the backscattering coefficients of the observed scene [2]. In these methods, all objects in the observed scene are treated as independent scatters and processed indiscriminately. When the scattering intensity of the target is relatively weak, the signal-to-noise ratio (SCR) of the reconstructed SAR image will be low. This is detrimental to the subsequent SAR image target detection, which generally refers to image preprocessing, feature extraction, and target localization [3], [4], [5], [6]. For example, roll-invariant features are investigated and extracted to detect targets from different orientations [6]. However, low SCR can compromise the effectiveness of target feature extraction, However, low SCR can affect the effectiveness of extracting target features, thereby reducing target detection performance. Therefore, these conventional SAR image reconstruction methods are not suitable for target detection in the case of strong clutter. If the target can be enhanced and clutter suppressed, the SCR of SAR images can be improved, which is beneficial for target detection.

Considering this requirement, feature-based SAR image reconstruction methods have been proposed by exploiting feature priors [7], [8], [9]. The commonly used feature priors include sparsity prior [7], regional continuity features [8], and edge features [9]. Among these methods, the sparsity-driven methods were proposed based on compressed sensing theory to reconstruct the strong scatters from the noisy and sparse samplings [10], [11]. Despite their effectiveness in reconstructing the strong scatters, sparsity-driven methods have a limitation in low SCR situations. When the scattering intensity of the clutter is stronger than the target, sparsity-driven methods may adversely affect target detection. To accommodate complex scenarios, methods using more sophisticated feature priors have also been proposed [12], [13]. For instance,

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Min Li is with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu 611756, China (e-mail: limin9@swjtu.edu.cn).

Weibo Huo, Junjie Wu, and Jianyu Yang are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China.

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Samadi et al. [12] propose to use mixed feature dictionaries to formulate SAR image reconstruction as a sparse representation problem, This method can simultaneously enhance multiple features. However, the performance of this method is constrained by the feature dictionaries. Another approach is to introduce regularization functions in SAR image reconstruction, which can overcome the limitation of feature dictionaries. In [13], joint sparsity and low-rankness regularizers are introduced, where the former models the sparse features and the latter represents the correlated homogeneous features. All in all, feature-based SAR image reconstruction methods have the potential to enhance specific features. However, their performance is limited by the use of handcrafted feature representations. Furthermore, high-resolution image reconstruction poses a greater challenge when the targets have more details.

Deep learning has the potential to overcome the limitation of handcrafted feature priors and help improve the adaptability of image reconstruction methods [14]. Recently, deep learning has gained great popularity in the image reconstruction field, such as ultrasound imaging [15]. Compared to conventional methods, deep learning-based methods have exhibited superior performance due to their ability to leverage data-driven feature learning. Deep learning has also been applied in SAR image reconstruction [16], [17], [18]. In [16], a convolutional neural network (CNN)-based autoencoder network is adopted to learn the image reconstruction and echo generation processes. This method can achieve improved sparsity-driven SAR imaging. However, CNNs can only learn local features and ignore global features with long-range correlation. Meanwhile, due to its black-box nature, the resulting image reconstruction network lacks interpretability. To improve interpretability, the deep unrolling algorithm is used in SAR image reconstruction [17], [18]. For example, Alver et al. [17] propose a SAR imaging network by unrolling an iterative solution, where the plug-and-play (PnP) prior is used to learn the structural feature priors. The results show that the deep prior has excellent feature learning capability and the unrolled network has good interpretability. In Li et al. [18], we proposed a SAR image reconstruction network that enhances targets using CNNs and a linear function to learn the target features. The results in [18] indicate that the deep network-based method can achieve target-enhanced SAR image reconstruction. However, the method in [18] uses only the target features and does not consider the clutter, which limits its application to complex clutter situations. Features of clutter, such as amplitude statistical features and spatial correlation, are also useful for target and clutter separation. Therefore, in this article, we further investigate how to perform SAR image reconstruction which is advantageous for subsequent target detection.

Considering target detection requirements, the proposed method utilizes the feature priors of both target and clutter deep to obtain SCR-enhanced SAR images. In particular, a self-attention CNN (SACNN)-based deep prior learning is proposed to exact the target and clutter feature priors. With the learned deep priors, the proposed method can separate targets from clutter during the image reconstruction process, which is advantageous for target detection. Experimental results show that the proposed method can significantly improve the SCRs of SAR images in the case of a strong clutter background. The main contributions of this article are as follows.

1) We propose a novel SAR image reconstruction method to exploit the deep priors of both the target and the clutter. Using these priors, we can better separate the target from the clutter during image reconstruction and help improve target detection performance.
2) We design a SACNN-based deep prior learning operation, which can extract local features by CNN and extract long-range global features by self-attention. The learned deep priors can effectively help separate the target from the clutter.
3) We construct an unrolled deep network to implement the image reconstruction and use a well-designed loss function. This implementation allows for deep prior learning and adaptive parameter optimization, ensuring the effectiveness and efficiency of the proposed solution.

This article is organized as follows. In Section II, we introduce the SAR signal model and image reconstruction problem model. Then the proposed deep network solution is given in Section III. In Section IV, the experimental results and analysis are shown. Finally, Section V provides the conclusions.

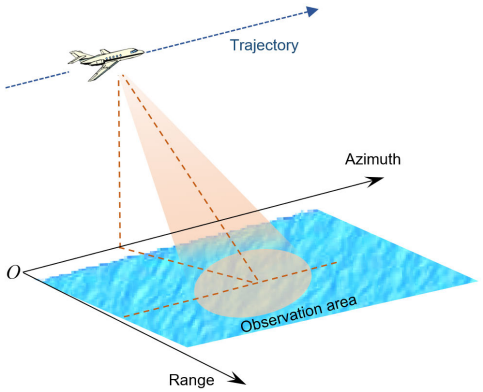


Fig. 1. Geometric configuration of SAR system.

II. Problem Modeling

This section introduces the SAR signal model and image reconstruction model. We assume the SAR system works in spotlight mode, and adopt the "stop-and-go" model to formulate the signals. Fig. 1 shows the geometric configuration of the SAR system.

A. Signal Model

Assuming SAR transmits the linear frequency modulation pulses, the sampled discrete echo can be represented as [2]

$$y[n,m]=\Sigma_{j=1}^{J}\left\{\begin{array}{c}\sigma_j\cdot w_a[m]\cdot w_r[n]\cdot\exp\{-j2\pi f_o n_{m,j}\}\\ \cdot\exp\{j\pi K_r(n-n_{m,j})^2\}\end{array}\right\} \tag{1}$$

where J represents the number of scatters, n the range sampling time, m the azimuth sampling time, σ_n the backscattering coefficient, $w_r[n]$ and $w_a[m]$ are the range and azimuth window functions, respectively, $n_{m,j}$ is the propagation time

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delay, K_r is the frequency modulation rate, f_0 is the carrier frequency, and c is the light speed.

By storing the data in matrix form, the discrete SAR signal model in (1) has the following expression:

$$\mathbf{Y} = \mathcal{G}\{\mathbf{X}\} + \mathbf{N} \tag{2}$$

where $\mathbf{Y} \in \mathbb{C}^{N\times M}$ represents the echo data, $\mathcal{G}\{\cdot\}$ is the echo generation process as illustrated in (1), $\mathbf{X} \in \mathbb{C}^{P\times Q}$ is the discrete SAR image, and $\mathbf{N} \in \mathbb{C}^{N\times M}$ is the white Gaussian noise. In essence, SAR image reconstruction aims to obtain $\mathbf{X}$ from $\mathbf{Y}$.

Considering the efficiency of data storage and process, this article adopts the approximate signal model proposed in [18]. As illustrated in [18], the MF-based approximate SAR echo generation operation $\mathcal{G}\{\cdot\}$ and the corresponding inverse operation $\mathcal{I}\{\cdot\}$ (image generation operation) are defined as follows:

$$G\{\cdot\} \triangleq \mathbf{F}_a^{-1}\left\{\mathbf{F}_r^{-1}\left[\mathbf{F}_r\left(\mathbf{F}_a(\cdot)\odot\mathbf{\Phi}_3^*\right)\odot\mathbf{\Phi}_2^*\right]\odot\mathbf{\Phi}_1^*\right\} \tag{3}$$

$$I\{\cdot\} \triangleq \mathbf{F}_a^{-1}\left\{\mathbf{F}_r^{-1}\left[\mathbf{F}_r\left(\mathbf{F}_a(\cdot)\odot\mathbf{\Phi}_1\right)\odot\mathbf{\Phi}_2\right]\odot\mathbf{\Phi}_3\right\} \tag{4}$$

where $\mathbf{F}_a(\cdot)$ and $\mathbf{F}_r(\cdot)$ represent the Fourier transform along the azimuth and range, respectively, while $\mathbf{F}_a^{-1}(\cdot)$ and $\mathbf{F}_r^{-1}(\cdot)$ represent the corresponding inverse Fourier transforms, $\mathbf{\Phi}_1$, $\mathbf{\Phi}_2$, and $\mathbf{\Phi}_3$ are the phases for range and azimuth compression, and $\odot$ denotes the Hadamard product operation.

B. Image Reconstruction Model

SAR image target detection is commonly understood as the extraction of the number and location of the targets from SAR images. Therefore, the ability to separate targets from clutter is a key determinant of target detection. Therefore, separating targets and clutter during the imaging process is of great help in improving target detection performance. Taking these into account, the formulated SAR image reconstruction model is presented as follows:

$$\begin{aligned}&\arg\min_{\mathbf{X},\mathbf{T},\mathbf{B}} \|\mathbf{Y}-\mathcal{I}\{\mathbf{X}\}\|_F^2 + \lambda_1\mathcal{R}_1(\mathbf{T}) + \lambda_2\mathcal{R}_2(\mathbf{B})\\ &\text{s.t. } \mathbf{X} = \mathbf{T}+\mathbf{B}\end{aligned} \tag{5}$$

where $\|\cdot\|_F^2$ is the Frobenius norm, $\mathbf{T}$ represents the target, $\mathbf{B}$ represents the clutter, $\mathcal{R}_1(\cdot)$ is the regularization function representing the target features, while $\mathcal{R}_2(\cdot)$ is the one representing the clutter features, and λ_1 and λ_2 are the corresponding coefficients.

In the proposed image reconstruction model (5), we aim to separate the targets from the clutter by introducing feature prior-based regularization functions. This approach can improve target detection performance by processing detect target from image $\mathbf{T}$. Therefore, solving the problem in (5) can lead to an image reconstruction result that is beneficial to subsequent target detection.

In practical applications, the explicit expression of the regularization functions in (5) is difficult to obtain. To deal with this issue, we use a SACNN-based deep prior learning operation to extract the priors of the target and clutter. The proposed deep prior learning operation can extract not only the local features but also the long-range global features. By optimizing the parameters in the deep prior learning, the feature priors represented by these regularization functions can be learned effectively in a data-driven manner.

III. Proposed Image Reconstruction Network

This section presents the deep network solution to the problem in (5). To construct the deep network in an interpretable manner, the deep unrolling algorithm [14] is adopted. The deep unrolling algorithm enables the construction of a deep network by unrolling an iterative solution. In the following, we first derive the ADMM-based iterative solution, and then construct the image reconstruction network.

A. Iterative Solution

From the expression in (5), we can see that the image reconstruction problem is a constrained multivariable optimization problem. We first use the augmented Lagrangian multiplier (ALM) algorithm to transform the constrained problem into an unconstrained one. According to ALM [19], we can obtain the Lagrangian function of (5) as follows:

$$\begin{aligned}\mathcal{L}(\mathbf{X},\mathbf{T},\mathbf{B},\mathbf{L}) = &\|\mathbf{Y}-\mathcal{I}\{\mathbf{X}\}\|_F^2 + \lambda_1\mathcal{R}_1(\mathbf{T})\\ &+ \lambda_2\mathcal{R}_2(\mathbf{B})\\ &+ \langle\mathbf{L},\mathbf{X}-\mathbf{T}-\mathbf{B}\rangle + \frac{\eta}{2}\|\mathbf{X}-\mathbf{T}-\mathbf{B}\|_F^2\end{aligned} \tag{6}$$

where $\mathbf{L}$ denotes the Lagrangian multiplier, η is the coefficient, and $\langle\cdot,\cdot\rangle$ is the inner product operation.

The minimization of the Lagrangian function in (6) is an unconstrained multi-variable optimization problem, and we can solve this problem by the ADMM algorithm [20]. According to the ADMM algorithm, the minimization of (6) can be obtained by iteratively updating the variables $(\mathbf{X},\mathbf{T},\mathbf{B},\mathbf{L})$ one by one, and the solutions for updating these variables are as follows:

$$\arg\min_{\mathbf{X}} \|\mathbf{Y}-\mathcal{I}\{\mathbf{X}\}\|_F^2 + \mathbf{\Pi}\left(\mathbf{X},\mathbf{T}^{(i)},\mathbf{B}^{(i)},\mathbf{L}^{(i)}\right) \tag{7a}$$

$$\arg\min_{\mathbf{T}} \lambda_1\mathcal{R}_1(\mathbf{T}) + \mathbf{\Pi}\left(\mathbf{X}^{(i+1)},\mathbf{T},\mathbf{B}^{(i)},\mathbf{L}^{(i)}\right) \tag{7b}$$

$$\arg\min_{\mathbf{B}} \lambda_2\mathcal{R}_2(\mathbf{B}) + \mathbf{\Pi}\left(\mathbf{X}^{(i+1)},\mathbf{T}^{(i+1)},\mathbf{B},\mathbf{L}^{(i)}\right) \tag{7c}$$

$$\mathbf{L}^{(i+1)} \leftarrow \tau_i\left(\mathbf{X}^{(i+1)}-\mathbf{T}^{(i+1)}-\mathbf{B}^{(t-1)}\right) + \mathbf{L}^{(i)} \tag{7d}$$

where τ_i denotes the step size for updating $\mathbf{L}$, and $\mathbf{\Pi}(\mathbf{X},\mathbf{T},\mathbf{B},\mathbf{L})$ is defined as

$$\begin{aligned}\mathbf{\Pi}(\mathbf{X},\mathbf{T},\mathbf{B},\mathbf{L}) &= \langle\mathbf{L},\mathbf{X}-\mathbf{T}-\mathbf{B}\rangle + \frac{\eta}{2}\|\mathbf{X}-\mathbf{T}-\mathbf{B}\|_F^2\\ &= \frac{\eta}{2}\left\|\mathbf{X}-\mathbf{T}-\mathbf{B}+\frac{1}{\eta}\mathbf{L}\right\|_F^2 - \frac{1}{2\eta}\|\mathbf{L}\|_F^2.\end{aligned} \tag{8}$$

Through solving the problems in (7), the updated variables $(\mathbf{X}^{(i+1)},\mathbf{T}^{(i+1)},\mathbf{B}^{(i+1)},\mathbf{L}^{(i+1)})$ can be calculated as follows.

1) Solution for $\mathbf{X}^{(i+1)}$: From (7a) and (8), the optimization for obtaining $\mathbf{X}^{(i+1)}$ is quadratic; hence (7a) can be solved analytically. The analytic solution of (7a) is given in (9), whose derivation process is provided in Appendix I

$$\mathbf{X}^{(i+1)} = \frac{1}{2+\eta}\left(2\mathcal{G}\{\mathbf{Y}\} + \eta\left(\mathbf{T}^{(i)}+\mathbf{B}^{(i)}\right) - \mathbf{L}^{(i)}\right) \tag{9}$$

where $\mathcal{G}\{\cdot\}$ is defined in (3).

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2) Solution for $\mathbf{T}^{(i+1)}$*:* From (7b) and (8), $\mathbf{T}^{(i+1)}$ can be calculated by

$$\arg\min_{\mathbf{T}} \frac{\eta}{2}\left\|\mathbf{T}-\mathbf{U}^{(i)}\right\|_F^2+\lambda_1\mathcal{R}_1(\mathbf{T}) \tag{10}$$

where $\mathbf{U}^{(i)}=\mathbf{B}^{(i)}-\mathbf{X}^{(i+1)}-\frac{1}{\eta}\mathbf{L}^{(i)}$.

Assuming the regularization function $\mathcal{R}_1(\cdot)$ and its proximal function are known, the iterative solution of (10) can be obtained via the accelerated proximal gradient (APG) method [19]. According to the APG algorithm, the iteration solution can be represented as

$$\mathbf{H}_k=\mathbf{T}_k^{(i+1)}+\varpi_k\left(\mathbf{T}_k^{(i+1)}-\mathbf{T}_{k-1}^{(i+1)}\right) \tag{11a}$$

$$\mathbf{T}_{k+1}^{(i+1)}=\text{prox}_{\gamma_k\mathcal{R}_1}\left(\gamma_k\mathbf{H}_k+(1-\gamma_k)\mathbf{U}^{(i)}\right) \tag{11b}$$

where ϖ_k and γ_k are the coefficients for the kth iteration, and $\text{prox}_{\gamma_k\mathcal{R}_1}(\cdot)$ represents the proximal function of $\mathcal{R}_1(\cdot)$ with coefficient γ_k.

In the iterative solution (11), the expression of the proximal function $\text{prox}_{\gamma\mathcal{R}_1}(\cdot)$ can not be obtained, as the regularization function $\mathcal{R}_1(\cdot)$ is not explicit. In Section III-B, this proximal function can be approximately implemented by the proposed deep prior learning operation.

3) Solution for $\mathbf{B}^{(i+1)}$*:* From (7c) and (8), $\mathbf{B}^{(i+1)}$ can be calculated by solving the following problem:

$$\arg\min_{\mathbf{B}} \frac{\eta}{2}\left\|\mathbf{B}-\mathbf{S}^{(i)}\right\|_F^2+\lambda_2\mathcal{R}_2(\mathbf{B}) \tag{12}$$

where $\mathbf{S}^{(i)}=\mathbf{T}^{(i+1)}-\mathbf{X}^{(i+1)}-\frac{1}{\eta}\mathbf{L}^{(i)}$.

Similar to the analysis in the solution for $\mathbf{T}^{(i+1)}$, the iterative solution of (12) can be obtained by the APG algorithm, and the iteration equations are

$$\mathbf{F}_k=\mathbf{B}_k^{(i+1)}+\upsilon_k\left(\mathbf{B}_k^{(i+1)}-\mathbf{B}_{k-1}^{(i+1)}\right) \tag{13a}$$

$$\mathbf{B}_{k+1}^{(i+1)}=\text{prox}_{\mu_k\mathcal{R}_2}\left(\mu_k\mathbf{F}_k+(1-\mu_k)\mathbf{S}^{(i)}\right) \tag{13b}$$

where υ_k and μ_k are the coefficients, and $\text{prox}_{\mu\mathcal{R}_2}(\cdot)$ is the proximal function of $\mathcal{R}_2(\cdot)$, which will be approximated by a deep network module in the proposed image reconstruction method.

4) Solution for $\mathbf{L}^{(i+1)}$*:* From (7d), the solution for $\mathbf{L}^{(i+1)}$ can be written as

$$\mathbf{L}^{(i+1)}=\tau_i\left(\mathbf{X}^{(i+1)}-\mathbf{T}^{(i+1)}-\mathbf{B}^{(i+1)}\right)+\mathbf{L}^{(i)}. \tag{14}$$

The above analysis shows that the proposed image reconstruction model in (5) can be solved iteratively, and the iterative solution can be summarized in Algorithm 1 as below.

Algorithm 1 Iterative Solution Based on ADMM&APG

INPUT: $\mathbf{Y}$, $\boldsymbol{\theta}$ $\mathcal{G}\{\cdot\}$, I, K, λ_1; λ_2, $\{\tau_i\}_{i=1}^{I}$, $\{\omega_k,\ \gamma_k,\ \upsilon_k,\ \mu_k\}_{k=1}^{K}$
Initialization $t=0$, $\mathbf{X}^{(0)}$, $\mathbf{T}^{(0)}$, $\mathbf{B}^{(0)}$, $\mathbf{L}^{(0)}$
  While $i<I$ **do**
  X-Update: Calculate $\mathbf{X}^{i+1}$ via (9)
  T-Update: Calculate $\mathbf{T}^{i+1}$
    Initialization $k=0$, $\mathbf{T}_0^{(i+1)}$
    While $k<K$ **do**
      Calculate $T^{(i+1)}$ via (11)
      $k=k+1$
    End while
    Return $\mathbf{T}^{(i+1)}=\mathbf{T}_K^{(i+1)}$
  B-Update: Calculate $B^{(i+1)}$
    Initialization $k=0$, $\mathbf{B}^{(i+1)}$
      While $k<K$ **do**
      Calculate $B_k^{(i+1)}$ via (13)
    $k=k+1$
    End while
    Return $\mathbf{B}^{(i+1)}=\mathbf{B}^{(i+1)_K}$
  L-Update: Calculate $\mathbf{L}^{(i+1)}$ via (14)
  $i=i+1$
End while
OUTPUT: $\mathbf{X}=\mathbf{X}(^{(I)}$, $\mathbf{T}=\mathbf{T}^I$

B. APG-SACNN-Net: Deep Network Solution

As discussed in Section II-B, the regularization functions in the proposed image reconstruction model represent the feature priors of the target and the clutter. However, obtaining the explicit expressions of these functions and their corresponding proximal functions is challenging in practice. To overcome this challenge, we approximate these proximal functions using a SACNN-based deep prior learning operation $\mathscr{F}\{\cdot\}$ in the proposed network. By unrolling the iterations of Algorithm 1, we can obtain the deep network solution to (5), named APG-SACNN-Net, whose network architecture is shown as follows.

1) Architecture of APG-SACNN-Net: Based on the deep unrolling algorithm, the APG-SACNN-Net can be constructed as Fig. 2 by taking each iteration in Algorithm 1 as a network layer. APG-SACNN-Net consists of multiple basic iteration blocks, and each iteration block consists of four calculation modules, namely X-Update, T-Update, B-Update, and L-Update. This design allows APG-SACNN-Net to be interpreted as an iterative solution, making it highly transparent in terms of its inner workings.

The input of APG-SACNN-Net is the sampled echo data $\mathbf{Y}$, and the output includes the scene image $\mathbf{X}$ and the target image $\mathbf{T}$. In the reconstructed target image $\mathbf{T}$, the target is enhanced while the clutter is suppressed. Hence, the target detection performance based on image $\mathbf{T}$ can be effectively improved.

In APG-SACNN-Net, the four calculation modules correspond to the four main parts in the iterative solution, whose structures are shown in the bottom half of Fig. 2. The X-Update and L-Update modules directly implement the calculations in (9) and (14), respectively. The T-Update and B-Update modules correspond to two iterative computation processes since the solutions given in (11) and (13) respectively. In Fig. 2, the blue line-arrows indicate the iteration loops. Moreover, the proximal functions prox$(\cdot)$ in (11) and (13) are approximated by the proposed deep prior learning operation $\mathscr{F}\{\cdot\}$, whose structure is given in Section III-B2.

2) Deep Prior Learning Operation: The regularization functions in (5) transform the inputs from the image domain to the feature domains, where the target and the clutter can be distinguished. Therefore, the regularization functions $\mathcal{R}_1(\cdot)$ and $\mathcal{R}_2(\cdot)$ in (5) can be interpreted as feature extraction

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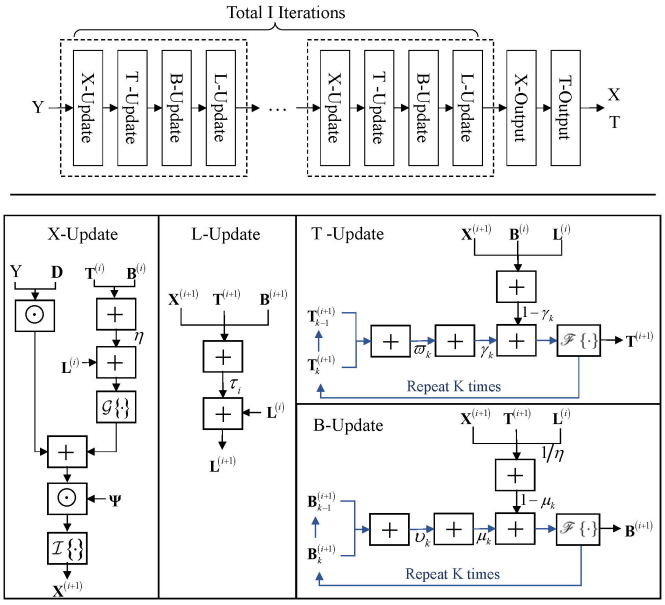


Fig. 2. Architecture of APG-SACNN-Net.

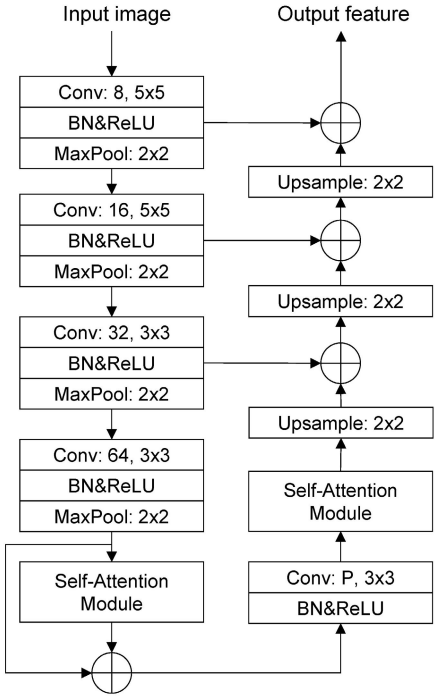


Fig. 3. Structure of the proposed deep prior learning operation $\mathscr{F}(\cdot)$ used in APG-SACNN-Net.

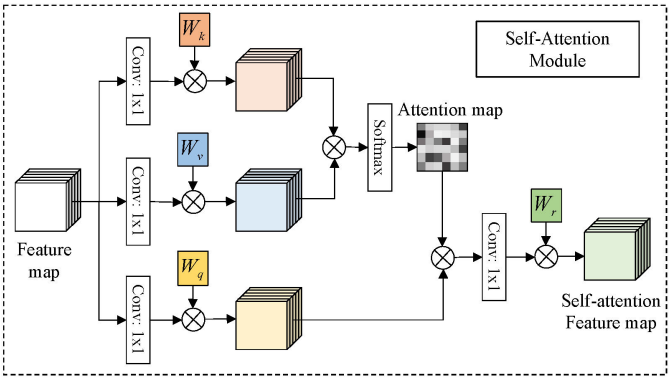


Fig. 4. Structure of self-attention module used in the proposed deep prior learning operation.

operations for the target and clutter respectively,

$$\text{prox}_{\lambda f}(v) = \underset{x}{\text{agrmin}}\, f(x) + \frac{1}{2\lambda}\|x - v\|_2^2. \tag{15}$$

The definition of the proximal function is shown in (15), which is a common optimization problem, particularly in regularized optimization [19]. From this definition, we can see that the proximal function $\text{prox}_{\lambda f}(\cdot)$ is completely determined by the original function $f(\cdot)$. Therefore, in the case of image reconstruction, the proximal function $\text{prox}_{\gamma\mathcal{R}_1}(\cdot)$ in (11) and $\text{prox}_{\mu\mathcal{R}_2}(\cdot)$ in (13) are related to the target features and clutter features, respectively. Based on this, we can approximate these proximal functions with feature extraction networks.

As shown in Figs. 3 and 4, we construct the deep prior learning operation $\mathscr{F}\{\cdot\}$ by CNN layers and self-attention module. In the proposed deep prior learning operation, the CNN layers are employed to extract local features, while the self-attention layers are utilized to extract global features with long-range correlation. In SAR image-based target detection applications, the target features are usually extracted within a small region, such as the attribute scattering center (ASC) feature and structure feature. The local ASC and structure features of the targets can be effectively extracted by the CNN layers [21]. Besides, as the targets to be detected are usually artificial objects, the learned ASC features can characterize the basic components of these objects, making the proposed method potential to be applied to various types of targets. Meanwhile, the clutter features, such as intensity distribution, texture, and correlation, are usually obtained globally. The self-attention module in Fig. 4 can obtain the long-range correlation and extract the global features [22]. Therefore, the proposed SACNN-based deep prior learning operation can extract both the local and global features of the target and the clutter. By optimizing the parameters in this operation, APG-SACNN-Net is more suitable for complex scenarios than the methods that rely on handcrafted feature priors.

3) Network Parameters and Loss Function: Based on the architectures of APG-SACNN-Net and the feature learning operation, the learnable parameters in APG-SACNN-Net can be given as

$$\mathcal{O} \triangleq \left\{\eta^t, \lambda_1^t, \lambda_2^t, \Omega_1^t, \Omega_2^t\right\}_{t=1}^{T} \tag{16}$$

where $\Omega_1^t \triangleq \{\varpi_k^t, \gamma_k^t, (\mathcal{C}_1, \mathcal{S}_1)_k^t\}_{k=1}^{K_1}$ represents the parameters of the feature learning operation in T-Update module, and $\Omega_2^t \triangleq \{\upsilon_k^t, \mu_k^t, (\mathcal{C}_2, \mathcal{S}_2)_k^t\}_{k=1}^{K_2}$ represents the parameters in B-Update module, where $\mathcal{C}_i$ denotes the learnable parameters in CNN layers, and $\mathcal{S}_i$ represents parameters in the self-attention module.

The loss function of APG-SACNN-Net can affect the reconstructed results. In the context of SAR image target detection, the structural and distribution characteristics of the target directly affect the detection performance, along with the distribution characteristics of the clutter. Hence, the proposed

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method adopts a loss function constructed using measures that reflect the reconstruction performance of these characteristics.

First, the Cross-Entropy (CE) is adopted to measure the reconstruction performance of SAR image distribution characteristics. CE is defined as [23]

$$\mathrm{CE}(p,q) = -\sum_i p_i \log(q_i). \tag{17}$$

In the target image, since the clutter occupies most of the observed scene, the ratio of positive samples (nonzero points) and negative samples (zero points) in the target image is highly unbalanced. Considering this issue, we adopt balanced cross-entropy (BCE) to measure the reconstruction performance of the target distribution, which is defined as [23]

$$\mathrm{BCE}(p,q) = -\sum_i \begin{bmatrix} \beta_i p_i \log(q_i) + (1-\beta_i) \\ \times (1-p_i)\log(1-q_i) \end{bmatrix} \tag{18}$$

where β is the weight and can be obtained by $\beta = 1 - \frac{p}{N_{\text{total}}}$, where N_{total} represents the total number of pixels in the image.

Then, dice loss is introduced to measure the structural and contour characteristics of the reconstructed target image, given as [24]

$$\mathrm{DL}(p,q) = 1 - \frac{2\sum_i p_i q_i + 1}{\sum_i (p_i + q_i) + 1}. \tag{19}$$

Combining BCE and DL, the combo loss (CL) can be used to measure the overall performance of target image reconstruction, which is defined as [25]

$$\mathrm{CL}(p,q) = \alpha \mathrm{BCE}(p,q) - (1-\alpha)\mathrm{DL}(p,q) \tag{20}$$

where α denotes the weight to balance the flexibility of dice loss and the smoothness of cross-entropy.

In conclusion, the loss function of APG-SACNN-Net is

$$\mathcal{L}oss(\mathcal{O}) = \mathrm{CE}\left(\hat{\mathbf{X}}, \mathbf{X}_g\right) + \kappa \mathrm{CL}\left(\hat{\mathbf{T}}, \mathbf{T}_l\right) \tag{21}$$

where κ represents the weight to balance the reconstruction performance of SAR image and target image, $\hat{\mathbf{X}}$ and $\hat{\mathbf{T}}$ represent the SAR image and target image reconstructed by APG-SACNN-Net, respectively, while $\mathbf{X}_l$ and $\mathbf{T}_l$ represent the corresponding ground truths.

IV. Experiments and Discussions

This section presents experimental results and observations to validate the performance of APG-SACNN-Net. First, the effectiveness of APG-SACNN-Net is evaluated. Then APG-SACNN-Net is compared with our previous work in [18] and two other feature-based methods. Finally, ablation experiments are performed to test the role of the target prior and clutter prior in APG-SACNN-Net. Simulated data are used in these experiments, permitting easy adjustment of experimental conditions. The simulated data is generated using an existing SAR image $\mathbf{X}$ as the observed scene. Echo $\mathbf{Y}$ is obtained according to the signal model presented in Section II-A.

In this article, the total number of stages I in APG-SACNN-Net is set to 6, and the iteration number K_1 of the T-Update module is set to 3, while K_2 of the B-Update module is set to 2. When training APG-SACNN-Net, we use the RMSprop optimizer to update the parameters. The experiments are conducted on a platform with an NVIDIA Tesla P100.

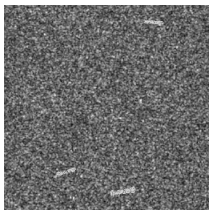


Fig. 5. Original image used to test the methods under different SNRs (Sentinel-1 sea surface SAR image).

In this article, normalized mean square error (NMSE), structural similarity (SSIM), and peak signal-to-noise ratio (PSNR) are adopted as performance measures. NMSE and PSNR are used to measure the image reconstruction accuracy, while SSIM measures the feature reconstruction performance. To evaluate the target reconstruction performance of the proposed method, the performance measures are calculated using the reconstructed target image $\hat{\mathbf{T}}$ and its corresponding ground truth $\mathbf{T}_l$. These measures are defined as follows:

$$\mathbf{NMSE} = \frac{\left\|\hat{\mathbf{T}} - \mathbf{T}_l\right\|_F^2}{\left\|\mathbf{T}_l\right\|_F^2} \tag{22}$$

where $\hat{\mathbf{T}}$ represents the reconstructed SAR image of targets and $\mathbf{T}_l$ denotes the ground truth

$$\mathbf{PSNR} = -10\log_{10}\left(\left\|\hat{\mathbf{T}} - \mathbf{T}_l\right\|_F^2 \Big/ N_{\mathbf{T}_l}\right) \tag{23}$$

where $N_{\mathbf{T}_l}$ denotes the total number of pixels in $\mathbf{T}_l$

$$\mathbf{SSIM} = \frac{\left(2\mu_{\hat{\mathbf{T}}}\mu_{\mathbf{T}_l} + C_1\right)\left(2\sigma_{\mu_{\hat{\mathbf{T}}}\mu_{\mathbf{T}_l}} + C_2\right)}{\left(\mu_{\hat{\mathbf{T}}}^2 + \mu_{\mathbf{T}_l}^2 + C_1\right)\left(\sigma_{\mu_{\hat{\mathbf{T}}}}^2 + \sigma_{\mu_{\mathbf{T}_l}}^2 + C_2\right)} \tag{24}$$

where $\mu_{\hat{\mathbf{T}}}$ and $\mu_{\mathbf{T}_l}$ represent the means, $\sigma_{\hat{\mathbf{T}}}$ and $\sigma_{\mathbf{T}_l}$ denote the standard deviations, $\sigma_{\mu_{\hat{\mathbf{T}}}\mu_{\mathbf{T}_l}}$ represents the cross-covariance, and C_1 and C_2 are parameters corresponding to the pixel variation range, which are set to $1e^{-4}$ and $9e^{-4}$ in our experiments.

A. Effectiveness of APG-SACNN-Net

This section presents the experiments conducted to verify the effectiveness of APG-SACNN-Net's different signal-to-noise ratios (SNRs), SCRs, and observed scenes.

1) Results with Different SNRs: To test the robustness of APG-SACNN-Net to noise, the noisy echo data are generated by adding Gaussian noise to $\mathbf{Y}$ in this experiment. The SAR image used to generate the testing data is given in Fig. 5, which comprises the sea surface and ships.

The image reconstruction results using the MF-based method at different SNRs are presented in Fig. 6, where the scene image $\hat{\mathbf{X}}$ and the target image $\hat{\mathbf{T}}$ reconstructed by APG-SACNN-Net are also shown. From these results, the reconstructed images exhibit more noise disturbances as the SNR decreases. Compared to the results of the MF-based method, the ship targets are more prominent in the images reconstructed by APG-SACNN-Net. In addition, APG-SACNN-Net can effectively recover ship features even

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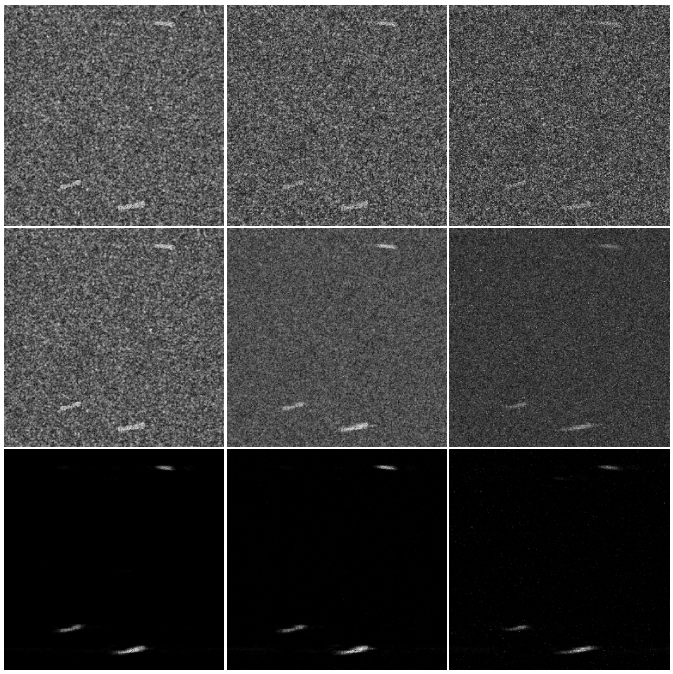


Fig. 6. SAR images reconstructed under different SNRs. From left to right: columns 1 to 3 show the images obtained under SNRs of 10, 5, and 1 dB, respectively. Row 1 shows the results of the MF-based method, row 2 shows the scene image $\hat{\mathbf{X}}$ reconstructed by APG-SACNN-Net, and row 3 shows the corresponding target image $\hat{\mathbf{T}}$.

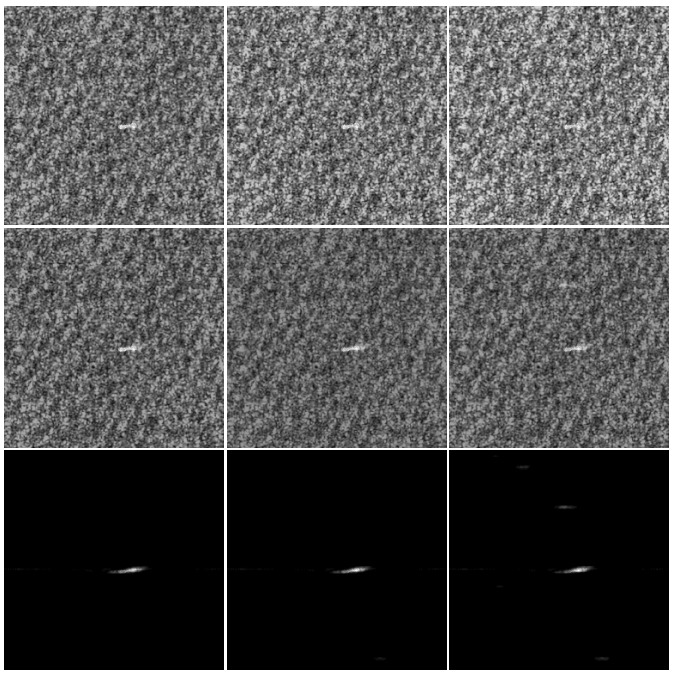


Fig. 7. SAR images reconstructed under different SCRs. From left to right: columns 1 to 3 show the images obtained under SNRs of 7, 5, and 3 dB, respectively. Row 1 shows the results of the MF-based method, row 2 shows the scene image $\hat{\mathbf{X}}$ reconstructed by APG-SACNN-Net, and row 3 shows the corresponding target image $\hat{\mathbf{T}}$.

TABLE I
PERFORMANCE MEASURES OF THE IMAGE RECONSTRUCTION RESULTS UNDER DIFFERENT SNRS

SNR	Method	NMSE↓	PSNR/dB↑	SSIM↑
10 dB	MF	0.009	20.43	0.69
	Proposed	**0.001**	**29.78**	**0.99**
5 dB	MF	0.025	15.84	0.48
	Proposed	**0.001**	**29.63**	**0.98**
1 dB	MF	0.059	12.03	0.25
	Proposed	**0.001**	**27.76**	**0.95**

with increased noise. For example, at an SNR of 1 dB, the ship targets are nearly indistinguishable from the clutter in the SAR image reconstructed by the MF-based method. By contrast, the three ship targets are clear in target image $\hat{\mathbf{T}}$ obtained by APG-SACNN-Net. In addition, the features of the targets are well recovered in $\hat{\mathbf{T}}$, which are useful for subsequent target detection.

Table I shows the average performance measures of 100 test experiments. Based on the NMSE and PSNR values presented in Table I, we can observe that APG-SACNN-Net demonstrates excellent image reconstruction performance and has more robust performance than the MF-based method. Furthermore, the SSIM values indicate that APG-SACNN-Net can effectively enhance the target features, which are advantageous for subsequent target detection.

2) Results with Different SCRs: The purpose of this experiment is to test the robustness of APG-SACNN-Net to different clutter levels, in particular, its ability to enhance target features under strong clutter conditions. The intensity of the clutter can be represented using SCR, which is defined as follows:

$$\mathbf{SCR} = 20\lg\left(\frac{|\mu_{\mathbf{B}} - \mu_{\mathbf{T}}|}{\sigma_{\mathbf{B}}}\right) \tag{25}$$

where $\mu_{\mathbf{T}}$ represents the mean intensity of the target area, $\mu_{\mathbf{B}}$ and $\sigma_{\mathbf{B}}$, respectively, denote the mean intensity and the standard deviation of the cluttered area.

The SAR images reconstructed by the MF-based method and APG-SACNN-Net under different SCRs (7, 5, and 3 dB) are given in Fig. 7. In the results of the MF-based method, the significance of the ship target in SAR images decreases as the SCR decreases. In comparison, the ship targets are more prominent in the images reconstructed by APG-SACNN-Net. When the SCR is 3 dB, the ship target can be easily distinguished from the clutter background in $\hat{\mathbf{X}}$ reconstructed by APG-SACNN-Net. In contrast, in the image obtained by the MF-based method, the strong cluttered background makes it difficult to distinguish the ship target in the scene. Moreover, APG-SACNN-Net can effectively obtain the target image $\hat{\mathbf{T}}$ and recover the features of the ship target with high quality. This is mainly because $\mathscr{F}(\cdot)$ can effectively learn the feature priors of the target and clutter. Therefore, APG-SACNN-Net can not only separate the target from the clutter but also enhance the target features that are beneficial for target detection.

In target detection applications, the SCR plays an important role in predicting detection performance. Typically, a higher SCR allows a better target detection performance. Hence, the SCR gain can be used as a useful index for image reconstruction methods to evaluate the effect of target enhancement.

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TABLE II
PERFORMANCE MEASURES OF THE IMAGE RECONSTRUCTION RESULTS UNDER DIFFERENT SCRS

SCR	NMSE↓	PSNR/dB↑	SSIM↑	G_{SCR}/dB↑
7 dB	0.001	28.69	0.97	24.64
5 dB	0.001	28.57	0.98	25.59
3 dB	0.002	28.03	0.96	25.16

To quantify the SCR gain of the proposed APG-SACNN-Net as compared to the MF-based method, we use $\mathbf{G_{SCR}}$ as an evaluation measure, which is defined as

$$\mathbf{G_{SCR}} = \mathbf{SCR}_{\hat{\mathbf{T}}} - \mathbf{SCR}_{\mathbf{X}_l} \tag{26}$$

where $\mathbf{SCR}_{\hat{\mathbf{T}}}$ and $\mathbf{SCR}_{\mathbf{X}_l}$ represent the SCR values of the target image $\hat{\mathbf{T}}$ and original SAR image $\mathbf{X}_l$, respectively.

Table II presents the average performance measures of images reconstructed by APG-SACNN-Net under different SCRs. These values are calculated by treating the image obtained by the MF-based method as the original image. The NMSEs, PSNRs, and SSIMs demonstrate the excellent image reconstruction and feature recovery performance of APG-SACNN-Net. Moreover, the $\mathbf{G_{SCR}}$ values indicate that APG-SACNN-Net can effectively enhance targets, which is advantageous for subsequent target detection.

3) Results with Different Observation Scenes: To examine the suitability of APG-SACNN-Net for different observed scenes, we use data generated by two distinct observed scenes: the sea surface scene and the land scene. This experiment aims to test the robustness and generality of APG-SACNN-Net. It should be noted that the training of APG-SACNN-Net is performed using Sentinel-1 sea surface SAR data and does not include land scenes in the training dataset.

Figs. 8 and 9, respectively, shows the images reconstructed by APG-SACNN-Net in sea surface scenes and land scenes. The original SAR images used to generate the echo data are displayed in the first row, followed by the reconstructed scene image $\hat{\mathbf{X}}$ and the target image $\hat{\mathbf{T}}$. From these results, APG-SACNN-Net exhibits excellent image reconstruction and target enhancement performance in both sea surface and land scenes. Although the vehicle targets in the land scene do not appear in the training data, APG-SACNN-Net is still able to separate the vehicle targets from the land clutter and reconstruct the vehicle target images with good quality. This is mainly attributed to the feature extraction capability of the proposed SACNN-based deep prior learning operation. As analyzed in Section III-B2, the SACNN can extract the ASC and structural features of the target, as well as the distribution and correlation features of the clutter. Therefore, the target and clutter features extracted by the proposed deep prior learning operation have a good generalization capability, enabling APG-SACNN-Net to be applied to different scenarios and target situations. In conclusion, these results demonstrate the excellent scene adaptation of APG-SACNN-Net.

The average performance measures of the APG-SACNN-Net reconstruction results in Figs. 8 and 9 are presented in Table III. The NMSEs, PSNRs, and SSIMs of the reconstructed results for these two scenes demonstrate that APG-SACNN-Net can achieve excellent reconstruction results for different observed scenes, indicating the robustness of APG-SACNN-Net. It is noteworthy that the $\mathbf{G_{SCR}}$ values

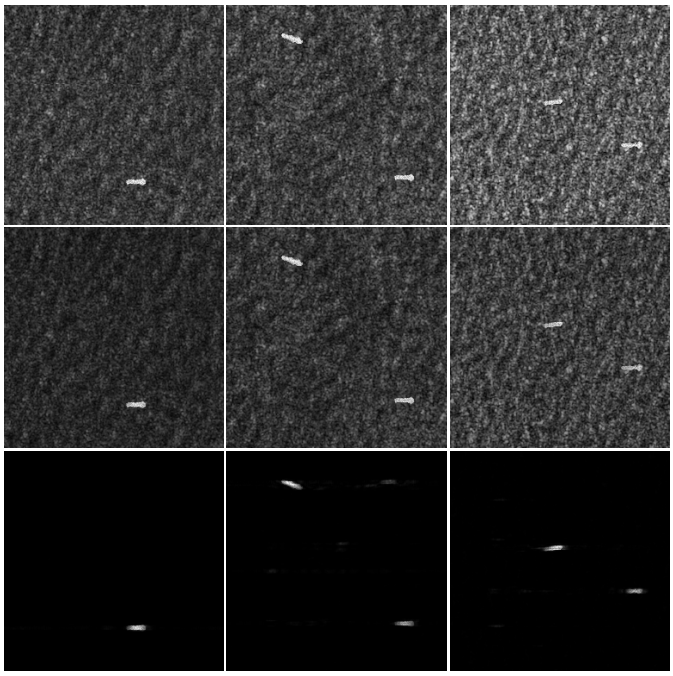


Fig. 8. Image reconstruction results with sea surface scenes (illuminated with different system configurations compared with the experiments in Sections IV-A1 and IV-A2). Columns 1 to 3 correspond to different sea surface scenes. Row 1 presents the original SAR images used to generate the echo data, row 2 and row 3 show the reconstructed scene image $\hat{\mathbf{X}}$ and target image $\hat{\mathbf{T}}$ of APG-SACNN-Net, respectively.

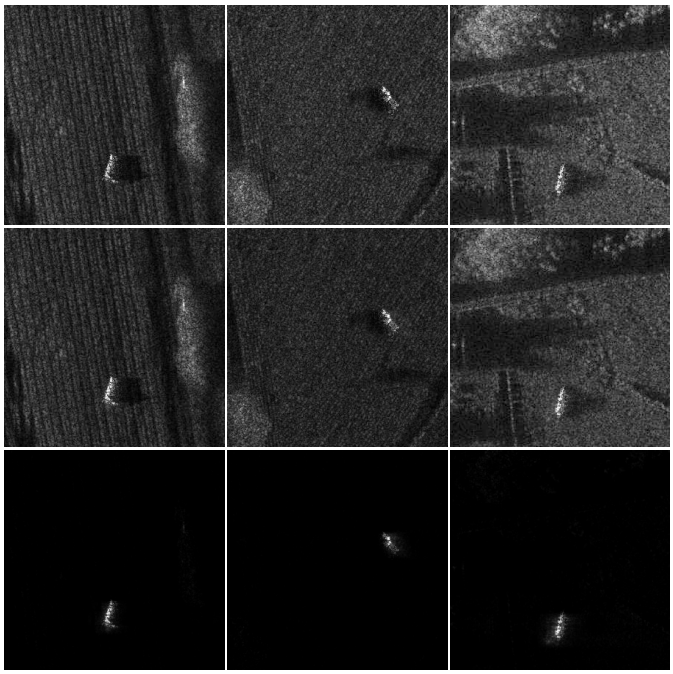


Fig. 9. Image reconstruction results with land scenes. Columns 1 to 3 correspond to three different land scenes. Row 1 presents the original SAR images used to generate the echo data, row 2 and row 3 show the reconstructed scene image $\hat{\mathbf{X}}$ and target image $\hat{\mathbf{T}}$ of APG-SACNN-Net.

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TABLE III
AVERAGE PERFORMANCE MEASURES OF RECONSTRUCTED IMAGES UNDER DIFFERENT OBSERVED SCENES

Conditions	NMSE↓	PSNR/dB↑	SSIM↑	G_{SCR}/dB↑
Sea Surface Scenes	0.001	30.281	0.981	23.45
Land Scenes	0.001	32.983	0.982	43.32

of these two scenes differ significantly due to the different SCR values of the original images. From these results, APG-SACNN-Net exhibits excellent target enhancement performance, suggesting that APG-SACNN-Net is advantageous in improving target detection performance in various scenes. Furthermore, during the training of APG-SACNN-Net, only the sea surface data are used as training data, and the land data in Fig. 9 are not included. Therefore, the results in Table III illustrate the generalisability of APG-SACNN-Net to the observed scenes and targets.

B. Comparisons With Target-Oriented Method

In this experiment, APG-SACNN-Net is compared with our previous work [18], which aims to enhance target features during image reconstruction using the target prior. Meanwhile, APG-SACNN-Net is also compared with another target-oriented SAR image reconstruction method [26], which utilizes the semantic information of targets through the Markov random field.

Fig. 10 shows the target reconstruction results for two types of targets. For the ship target experiment, the sea clutter is more complex than the one used in [18]. From the results in Fig. 10, the ship targets image obtained by [26] and [18] are contaminated by clutter residual, while APG-SACNN-Net can reconstruct the ship target image with high quality. This is because [26] will consider the continuous clutter area as the target area, and [18] does not use clutter features during the reconstruction process. When the clutter has complex properties, only using target features is not sufficient to separate the target from the clutter. For the car target experiment, APG-SACNN-Net and [26] outperform [18] in target image reconstruction. Since the car target is not included in the training dataset, the method in [18] cannot effectively extract car features using CNNs trained on ship targets. In contrast, in this article, the proposed deep prior learning operation uses SACNN to enable the learning of intrinsic target features (e.g., ASC) to improve the generalization capability of APG-SACNN-Net. Meanwhile, compared to [26], APG-SACNN-Net does not require parameter tuning and can adaptively achieve target reconstruction.

Table IV gives the average performance measures of the three methods. From NMSE and PSNR values, APG-SACNN-Net is more effective in target image reconstruction. In addition, the SSIM values show that in terms of target feature enhancement, APG-SACNN-Net is superior to [26] and [18] in both complex sea scene and land scene, indicating better robustness of APG-SACNN-Net.

Based on the above analysis, it can be concluded that APG-SACNN-Net outperforms the method in [18] and [26] in terms of clutter suppression and target enhancement due to the use of clutter features and SACNN-based feature learning.

TABLE IV
PERFORMANCE MEASURES OF THE TARGET RECONSTRUCTION RESULTS

Target	Method	NMSE↓	PSNR/dB↑	SSIM↑
Ship	Method in [26]	0.0012	29.29	0.88
	Method in [18]	0.0012	29.24	0.50
	Proposed	**0.0005**	**32.46**	**0.92**
Car	Method in [26]	0.0033	24.78	0.92
	Method in [18]	0.0043	23.64	0.65
	Proposed	**0.0032**	**24.96**	**0.93**

C. Comparisons With Other Methods

In the following experiments, we compare APG-SACNN-Net with a target-oriented SAR image reconstruction method (target-oriented) [26] and two feature-based methods, including an iterative method L1-CAMP [27] and a deep network method AF-AMPNet [28]. We compare the performance of image reconstruction and the effect of the reconstructed SAR images on subsequent target detection.

1) Image Reconstruction Performance: To better compare the reconstruction performance of these methods, various observed scene data are used in this experiment, including sea surface scenes and land scenes under noisy and strong clutter conditions. The SAR images reconstructed by APG-SACNN-Net and the comparison methods are demonstrated in Fig. 11, while the corresponding performance measures (average of 100 experiments) are demonstrated in Table V.

The images shown in Fig. 11 indicate that the L1-CAMP method and the AF-AMPNet method can suppress the weakly scattering clutter while retaining the strongly scattering target, resulting in sparse SAR images. The images in column 3 show that the target-oriented method has better target reconstruction and clutter suppression performance. However, in the SAR images reconstructed by these two methods, the targets consist of several discrete scatters, destroying the structural features of the targets, which is disadvantageous for subsequent target detection. In comparison, the proposed APG-SACNN-Net can reconstruct not only the whole observed scene but also the target image. Benefiting from the feature extraction capability of SACNN-based deep prior learning operation, APG-SACNN-Net can effectively separate the target and clutter to achieve target enhancement and clutter suppression. Moreover, the structural features of the targets are well recovered in the image $\hat{\mathbf{T}}$ reconstructed by APG-SACNN-Net, which is advantageous for the subsequent SAR image target detection.

Table V presents the average performance measures of the images shown in Fig. 11. From Table V, the target-oriented, L1-CAMP, and AF-AMPNet methods exhibit satisfactory image reconstruction performance, although they have slightly lower NMSE and PSNR values compared to the proposed APG-SACNN-Net. However, in terms of target enhancement and clutter suppression, as quantified by SSIM and SCR gain ($\mathbf{G_{SCR}}$), APG-SACNN-Net significantly outperforms the com-

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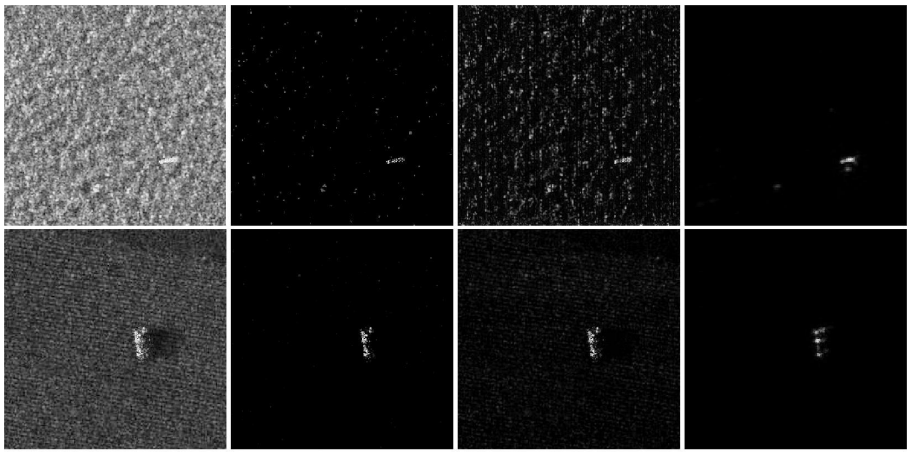


Fig. 10. Comparison of target reconstruction results. Row 1 and row 2 show the target reconstruction results of the ship target and car target, respectively. Column 1 shows the original SAR scenes, column 2 shows the target image reconstructed by the method in [26], column 3 shows the target image reconstructed by the method in [18], and column 4 shows the target image obtained by APG-SACNN-Net.

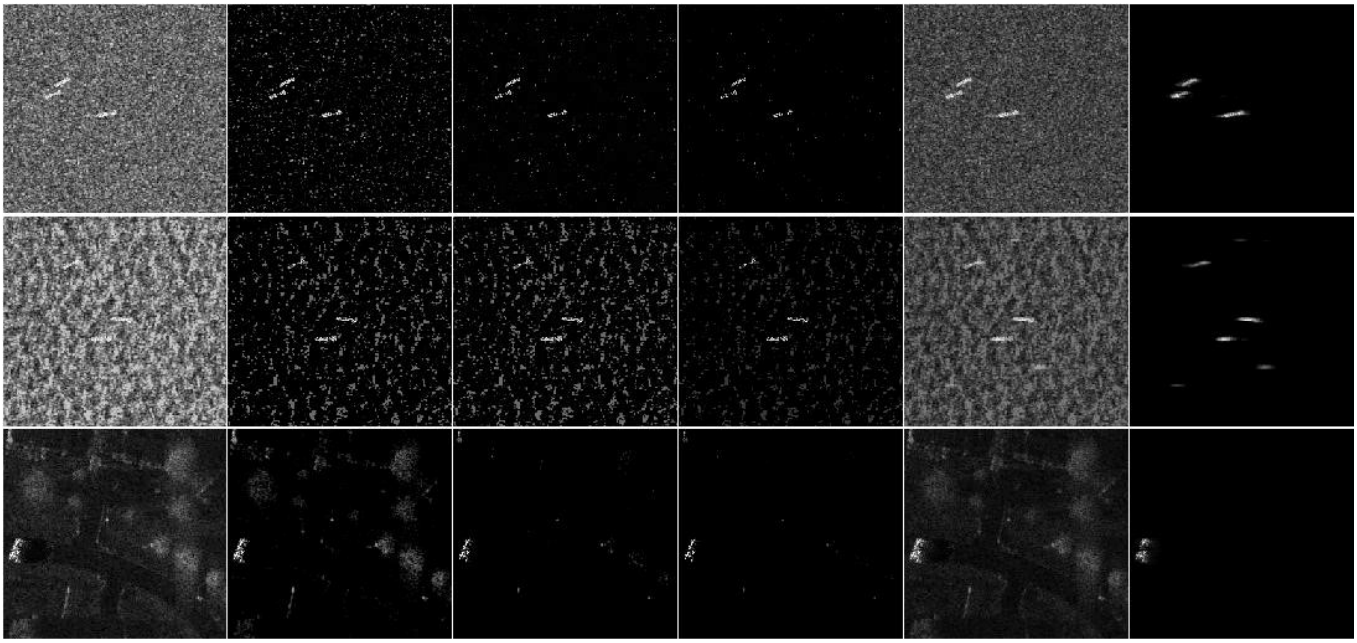


Fig. 11. SAR images reconstructed by different methods. Rows 1 to 3 correspond to three different observed scenes. Column 1 shows the original SAR images. Columns 2 shows the reconstructed SAR images of L1-CAMP, column 3 shows the images of AF-AMPNet, and column 4 shows the results of the target-oriented method. Column 5 and column 6 show the SAR image $\hat{\mathbf{X}}$ and target image $\hat{\mathbf{T}}$ reconstructed by APG-SACNN-Net.

TABLE V
PERFORMANCE MEASURES OF THE RECONSTRUCTED IMAGES OF DIFFERENT METHODS

Methods	NMSE↓	PSNR/dB↑	SSIM↑	$\mathbf{G_{SCR}}$/dB↑
L1-CAMP	0.006	21.999	0.253	6.98
AF-AMPNet	0.002	28.094	0.503	14.27
Target-Oriented	0.002	27.61	0.777	19.36
Proposed	**0.001**	**30.408**	**0.983**	**22.58**

parison methods. Specifically, The SSIM values demonstrate the superior feature reconstruction performance of APG-SACNN-Net, and the $\mathbf{G_{SCR}}$ values indicate the excellent separation of the target and clutter. These findings suggest that the image reconstruction results of APG-SACNN-Net are more conducive to subsequent target detection processing.

2) Target Detection Performance Improvement: To test the influence of SAR images reconstructed by different methods on target detection, we use a constant false alarm rate (CFAR) method [29] to perform SAR image target detection. It should be noted that the purpose of APG-SACNN-Net is to perform target-enhanced and clutter-suppressed SAR image reconstruction. Therefore, to better illustrate the effect of APG-SACNN-Net on target detection, we use CFAR for target detection instead of more sophisticated deep network-based detection methods. When performing target detection, the target image $\hat{T}$ reconstructed by APG-SACNN-Net is used as the input to CFAR, and the scene image $\hat{X}$ is used to display the results.

Fig. 12 shows the target detection results, where the green rectangular boxes indicate the correctly detected targets (TP), the red rectangular boxes indicate the wrongly detected targets

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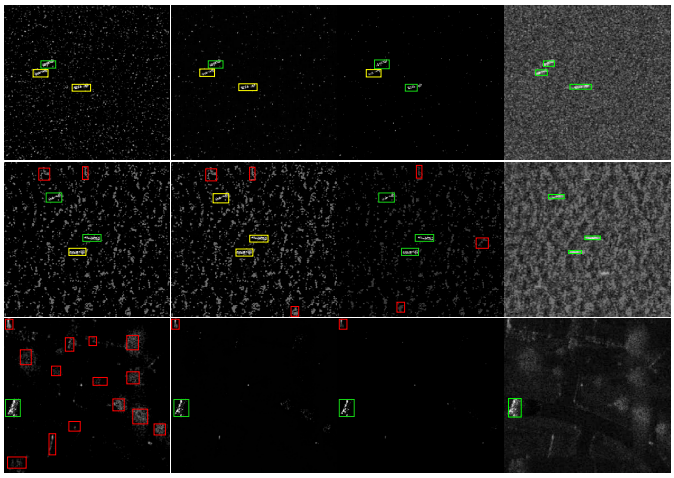


Fig. 12. Target detection results based on the reconstructed SAR images in Fig. 11. Row 1 to row 3 are three different observed scenes. Columns 1 to 4 show the detection results from the SAR images obtained by the L1-CAMP method, AF-AMPNet method, target-oriented method, and APG-SACNN-Net, respectively.

(FP), and the yellow rectangular boxes indicate the missed targets (FN). From the detection results, there are serious false alarms and missed targets in the results based on SAR images reconstructed by the target-oriented, L1-CAMP, and AF-AMPNet. This is mainly due to the inadequate feature extraction capabilities of these two methods, making them ineffective in recovering the target features and extracting the targets. In the results of the target-oriented method, the target detection rate has improved. However, the false alarm rate remains high due to background clutter in the images. In contrast, APG-SACNN-Net can effectively extract targets and recover the structural features of the targets. This is mainly a benefit from the excellent feature extraction capability of SACNN-based deep prior learning operation $\mathscr{F}(\cdot)$. Therefore, the image reconstruction results of APG-SACNN-Net are more advantageous for target detection. In Fig. 12, the targets are correctly detected from the image reconstructed by APG-SACNN-Net, showing its superiority for target detection.

To quantitatively evaluate the target detection performance, two measures, precision (F_{precise}) and recall (F_{recall}), are used in this experiment. Their definitions are given as follows:

$$F_{\text{precise}} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{27a}$$

$$F_{\text{recall}} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{27b}$$

where TP, FP, and FN denotes the numbers of correct detection, false alarm, and missed detection, respectively. From the definitions, F_{precise} measures the accuracy of the target detection results, and F_{recall} measures the ability to detect all targets.

Table VI shows the average performance measures of the target detection results. In Table VI, the time items demonstrate the image reconstruction time and CFAR target detection time, respectively. From F_{precise} values, the detection results corresponding to APG-SACNN-Net exhibit the best accuracy, while the L1-CAMP method has the worst accuracy.

TABLE VI

PERFORMANCE MEASURES FOR THE TARGET DETECTION PERFORMANCE

Methods	$\mathbf{F}_{\text{precise}} \uparrow$	$\mathbf{F}_{\text{recall}} \uparrow$	Time/s$\downarrow$
L1-CAMP	75.03%	97.64%	$0.157 + 0.446$
AF-AMPNet	92.98%	92.60%	$\mathbf{0.001 + 0.439}$
Target-Oriented	93.61%	98.57%	$0.232 + 0.447$
Proposed	**96.90%**	**98.88%**	$0.002 + 0.443$

TABLE VII

NETWORK CONFIGURATION FOR ABLATION EXPERIMENT

Notation	Ablation Network Configuration
Abla_C	Ablate the prior learning of clutter in APG-SACNN-Net, and obtain clutter $\mathbf{B}$ by subtracting the target $\mathbf{T}$ from $\mathbf{X}$.
Abla_T	Ablate the prior learning of target in APG-SACNN-Net, and obtain target $\mathbf{T}$ by subtracting the clutter $\mathbf{B}$ from $\mathbf{X}$.

As shown in Fig. 12, more false alarms exist in the detection results corresponding to L1-CAMP, leading to a decrease in target detection accuracy. At the same time, the F_{recall} values indicate that all targets can be effectively detected from the target images reconstructed by APG-SACNN-Net. In contrast, in Fig. 12, the AF-AMPNet method has the most missed detection results, resulting in the worst performance of AF-AMPNet in terms of F_{recall}. For the target-oriented method, the target detection ability indicated by F_{recall} is greatly improved compared to the other two comparison methods. However, the false alarms caused by residual clutter degrade the target detection accuracy indicated by F_{precise}. Based on the analysis, we can conclude that the SAR images reconstructed by APG-SACNN-Net are more useful for improving the target detection performance.

In terms of computational time, the target detection processing times corresponding to these three methods are comparable, with a difference of less than 1.6%. As for the image reconstruction time, the AF-AMPNet method has the shortest processing time. APG-SACNN-Net is comparable to AF-AMPNet in terms of computational speed and is 116 times faster than the target-oriented method. Therefore, we can conclude that APG-SACNN-Net is computationally efficient.

D. Ablation Experiment

In this experiment, we will perform ablation experiments on the target prior and clutter prior modules respectively, so as to test the effect of target and clutter priors in APG-SACNN-Net. The network configuration for the ablation experiment is demonstrated in Table VII. We will compare the target reconstruction results of Abla_C, Abla_T, and APG-SACNN-Net under the same experimental conditions.

The results of the ablation experiments are shown in Fig. 13, where four different scenes are used to test the ablation networks and APG-SACNN-Net. In the results of Abla_C, the clutters are removed from the target images. However, the target reconstruction performance is severely degraded and even false targets are observed. This is because Abla_C

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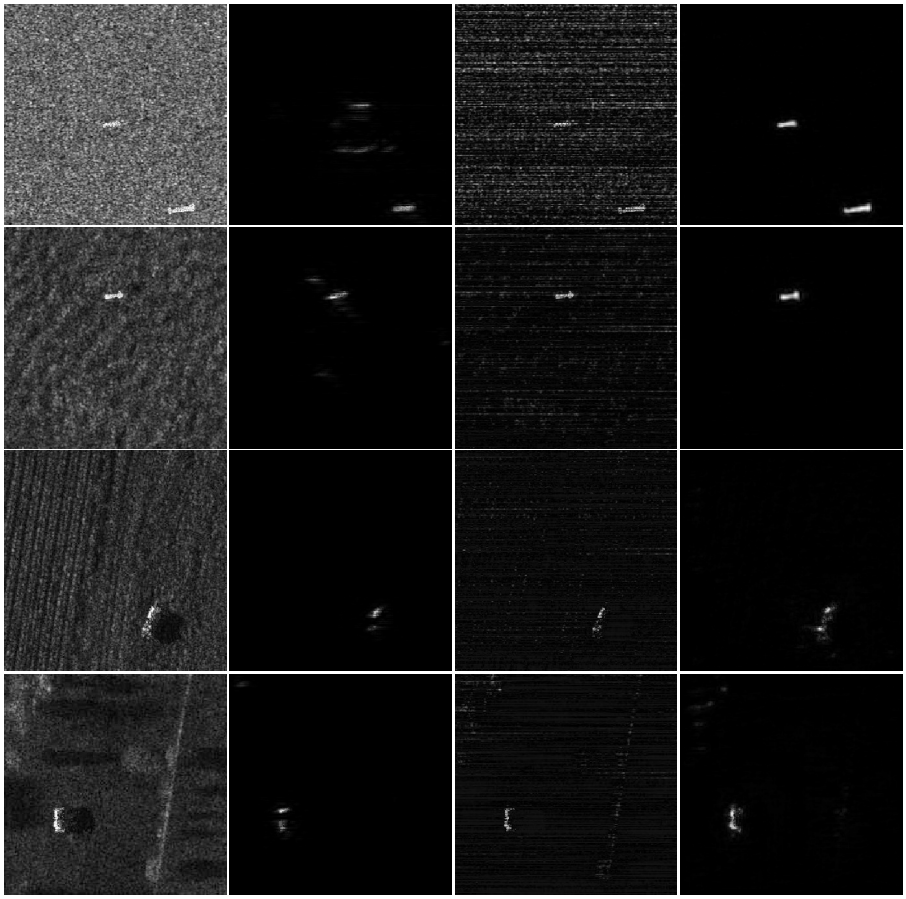


Fig. 13. Results of ablation experiments. Rows 1 to 4 represent different scenes. Column 1 shows the original scene, column 2 shows the results of the MF-based method, and columns 3 to 4 represent the target reconstruction results of Abla_C, Abla_T, and APG-SACNN-Net.

only uses the target prior. When separating the target from the clutter, the strong clutter part may be misidentified as the target, which not only affects the performance of target reconstruction but also results in false targets. In the results of Abla_T, its target reconstruction performance is better than that of Abla_C. However, there is a large amount of clutter residual in the target images of Abla_T. This is because only the clutter prior is used in Abla_T. When reconstructing the target image, Abla_T will consider the noise in the clutter feature space as the target part, resulting in redundant clutter residues in the target image. In the target images of APG-SACNN-Net, not only the clutter is effectively removed, but also the features of the target are effectively reconstructed. From the above results and analysis, both target prior and clutter prior can help to achieve the separation of target and clutter during image reconstruction. However, only the target prior or the clutter prior is not sufficient to separate the target from the clutter and obtain target images.

Table VIII demonstrates the metric values of the results in Fig. 13. The values of NMSE and PSNR indicate that Abla_T outperforms Abla_C in the overall performance of target image reconstruction. Combined with the results in Fig. 13, it can be seen that the clutter prior plays a more significant role than the target prior in separating the target from the clutter. The values of SSIM show that Abla_C outperforms Abla_T in target feature reconstruction, which indicates the target prior plays a more significant role than the clutter prior in reconstructing target features. Therefore, APG-SACNN-Net, which uses both target prior and clutter prior, performs best in target-clutter separation and target feature reconstruction. The results in Fig. 13 and the metric values in Table VIII verify the feasibility and effectiveness of APG-SACNN-Net.

TABLE VIII
PERFORMANCE MEASURES FOR ABLATION EXPERIMENT RESULTS

Network	NMSE↓	PSNR/dB↑	SSIM↑
Abla_C	0.015	19.02	0.86
Abla_T	0.003	26.02	0.14
Proposed	**0.001**	**30.30**	**0.92**

V. CONCLUSION

This article presents an SAR image reconstruction network for target detection. To achieve target enhancement and clutter suppression, a SACNN-based deep prior learning operation is proposed. SACNN can extract both the target and clutter features. By integrating this learning operation, the proposed method can achieve target-enhanced and clutter-suppressed SAR image reconstruction. In addition, a deep unrolled network called APG-SACNN-Net is

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designed to enable data-driven deep priors learning. Experiments under different SNRs, SCRs, and observed scenes validate the image reconstruction and target enhancement performance of APG-SACNN-Net. Meanwhile, by comparing with other feature-based SAR image reconstruction methods, the results of APG-SACNN-Net are more advantageous for subsequent target detection. Furthermore, comparisons with target-oriented method and ablation experiments illustrate the effectiveness and superiority of APG-SACNN-Net. In future work, we will evaluate the proposed method with real data, and explore adaptability to more complex scenes.

APPENDIX I

This section presents the derivation of the solution of (7a). By combining (7a) with (8), the calculation of $\mathbf{X}^{(i+1)}$ can be achived by solving the problem below

$$\arg\min_{\mathbf{X}} \|\mathbf{Y}-\mathcal{I}\{\mathbf{X}\}\|_F^2+\frac{\eta}{2}\left\|\mathbf{X}-\mathbf{T}^{(i)}-\mathbf{B}^{(i)}-\frac{1}{\eta}\mathbf{L}^{(i)}\right\|_F^2. \quad (28)$$

For the convenience of the following derivation, the signal model in (2) is rewritten in the vector form as

$$\mathbf{y}=\mathbf{A}\mathbf{x}+\mathbf{n} \quad (29)$$

where $\mathbf{y}$, $\mathbf{x}$, and $\mathbf{n}$ respectively represent the vector forms of $\mathbf{Y}$, $\mathbf{X}$, and $\mathbf{N}$ in (2), and $\mathbf{A}$ is the measurement matrix, whose explicit expressions can be found in [18]. It should be noted that the measurement matrix $\mathbf{A}$ in (29) is a unitary matrix.

By representing the data in a vector form, the problem in (28) can be rewritten as

$$\arg\min_{\mathbf{x}} \|\mathbf{y}-\mathbf{A}\mathbf{x}\|_2^2+\frac{\eta}{2}\left\|\mathbf{x}-\mathbf{t}-\mathbf{b}+\frac{1}{\eta}\mathbf{l}\right\|_2^2 \quad (30)$$

where $\mathbf{t}$, $\mathbf{b}$, and $\mathbf{l}$ are the corresponding vector forms of $\mathbf{T}$, $\mathbf{B}$, and $\mathbf{L}$. The gradient of (30) with respect to $\mathbf{x}$ can be calculated as

$$\begin{aligned}\nabla J_{\mathbf{x}} &= 2\mathbf{A}^H(\mathbf{A}\mathbf{x}-\mathbf{y})+\eta\left(\mathbf{x}-\mathbf{t}-\mathbf{b}+\frac{1}{\eta}\mathbf{l}\right)\\ &= (2\mathbf{\Psi}+\eta\mathbf{I})\mathbf{x}-\left(2\mathbf{A}^H\mathbf{y}+\eta(\mathbf{t}+\mathbf{b})-\mathbf{l}\right)\end{aligned} \quad (31)$$

where $\mathbf{\Psi}=\mathbf{A}^H\mathbf{A}$, and $\mathbf{I}$ denotes the identity matrix.

By setting the gradient $J_{\mathbf{x}}$ equal to 0, the solution of $\mathbf{x}$ can be represented as follows:

$$\mathbf{x}=(2\mathbf{\Psi}+\eta\mathbf{I})^{-1}\left(2\mathbf{A}^H\mathbf{y}+\eta(\mathbf{t}+\mathbf{b})-\mathbf{l}\right). \quad (32)$$

Finally, by transforming the solution in (32) to its corresponding matrix form, we can obtain

$$\mathbf{X}=\frac{1}{2+\eta}(2\mathcal{G}\{\mathbf{Y}\}+\eta(\mathbf{T}+\mathbf{B})-\mathbf{L}) \quad (33)$$

where $\mathcal{G}\{\cdot\}$ represents the echo generation operation defined in (3).

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Min Li (Student Member, IEEE) received the B.S., M.S., and Ph.D. degrees in information and communication engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2013, 2016, and 2023, respectively.

She is currently a Lecturer with the School of Computer and Artificial Intelligence, Southwest Jiaotong University, Chengdu. Her research interests include SAR image reconstruction, machine learning, optimization, compressed sensing, and inverse problems.



Weibo Huo (Member, IEEE) received the B.S. degree from the School of Science, Shandong Jianzhu University, Jinan, China, in 2010, and the Ph.D. degree from the School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2019.

He is currently an Associate Research Fellow with the School of Information and Communication Engineering, UESTC. His research interests include statistical signal processing, machine learning, radar target detection, and sea clutter modeling and simulation.



Junjie Wu (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2004, 2007, and 2013, respectively.

From January 2012 to January 2013, he was a Visiting Student with the Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA. He is currently a Professor at UESTC. His research interests include synthetic aperture radar imaging (particular emphasis on bistatic synthetic aperture radar).

Dr. Wu is a reviewer of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON SIGNAL PROCESSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN SIGNAL PROCESSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *IET Radar, Sonar and Navigation*.



Jianyu Yang (Member, IEEE) received the B.S. degree in electronic engineering from the National University of Defense Technology, Changsha, China, in 1984, and the M.S. and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1987 and 1991, respectively.

He is currently a Professor with the School of Information and Communication Engineering, UESTC. His research interests are mainly in SAR and statistical signal processing.

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